DESIGN AND IMPLEMENTATION OF AN IOT-ENABLED FLOOD DETECTCTION AND NOTIFICATION SYSTEM

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https://youtu.be/1iN-ZQ9X3Os

ABSTRACT

Floods are one of the most frequent natural disasters, causing severe damage to life and property. This project presents an IoT-based flood detection and notification system that provides real-time monitoring of water levels and early warning alerts. The system uses an ultrasonic sensor to measure water levels, which are processed using an Arduino microcontroller. When the water level exceeds a predefined threshold, alerts are generated through a buzzer, LED indicators, and mobile notifications via the Telegram application using a Wi-Fi module.

The system is designed to be low-cost, efficient, and easy to deploy in flood-prone areas. It enables real-time monitoring and quick response, reducing the risk of damage and improving disaster preparedness. The proposed solution offers a reliable approach for early flood detection and can be further enhanced with cloud storage and predictive analytics.

TEAM MEMBER DETAILS

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INTRODUCTION

Flooding is one of the most frequent and destructive natural disasters in India, especially during the monsoon season. Heavy rainfall, poor drainage systems, and rapid urbanization often result in severe flooding in cities like Chennai and Mumbai. These floods cause significant loss of life, damage to infrastructure, and disruption of daily activities. Additionally, climate change has increased the intensity and unpredictability of rainfall, making flood situations more critical and harder to manage.

Traditional flood monitoring systems mainly depend on manual observation, historical data, and delayed communication methods such as radio or television alerts. These approaches are often inefficient as they do not provide real-time or localized warnings. As a result, people and authorities may not receive timely information to take preventive measures, leading to increased damage and risk during flood situations.

With the advancement of Internet of Things (IoT) technology, it is now possible to develop smart systems for real-time monitoring and early warning. This project focuses on designing an IoT-based flood detection and notification system that continuously monitors water levels using sensors and processes the data through a microcontroller. When the water level exceeds a predefined threshold, the system generates instant alerts through LED indicators, a buzzer, and mobile notifications, enabling quick response and improved disaster management.

METHODOLOGIES

The system follows a threshold-based detection approach using IoT components:

- Water level is measured using an **ultrasonic sensor**
- Data is processed using an **Arduino microcontroller**
- Threshold values are set to classify:
  - Safe
  - Warning
  - Danger levels
- Alerts are generated using:
  - LED indicators (Green, Blue, Red)
  - Buzzer for sound alert
  - Telegram notifications via Wi-Fi module (ESP8266)

The system continuously monitors water levels and triggers alerts when thresholds are exceeded.

ARCHITECTURE DIAGRAM

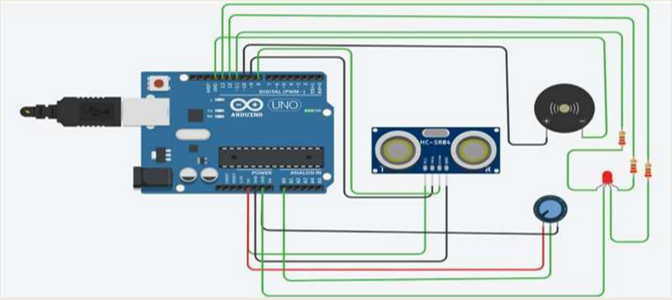

Table 1. Water Level vs Alert Status

Water Level (cm)	Status	LED Colour	Alert Type
5–10 cm	Safe	Green	No Alert
11–20 cm	Warning	Blue	Warning Alert
>20 cm	Danger	Red	Buzzer + Telegram

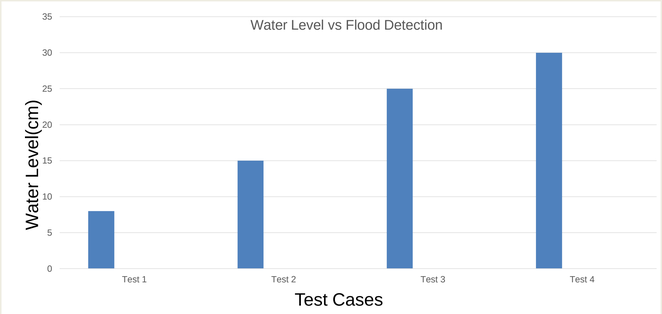

Chart 1. Water Level vs Flood Alert Status

RESULTS

- The system successfully detected changes in water levels in real-time
- Alerts were generated within a few seconds
- LED indicators correctly showed:
  - Green → Safe
  - Blue → Warning
  - Red → Danger
- Telegram notifications were received instantly

The system demonstrated reliable performance during testing under different water level conditions.

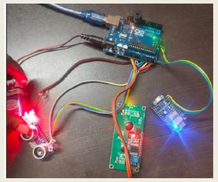

Figure 1. Red LED indicating danger

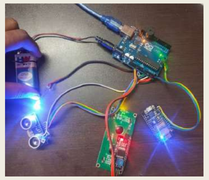

Figure 2. Blue LED indicating warning

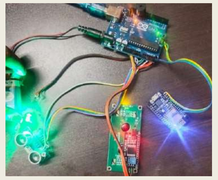

Figure 3. Green LED indicating safe

CONCLUSIONS

The project successfully developed an IoT-based flood detection and alert system that provides real-time monitoring and early warnings. The system is cost-effective, easy to implement, and reliable for flood-prone areas.

It helps reduce risks by providing timely alerts through multiple channels such as buzzer, LED, and mobile notifications. This project demonstrates the effective use of IoT technology in disaster management and public safety.

PUBLICATION

Presented in ICICCS 2026 (Erode, India) – Paper titled "IoT-Enabled Flood Detection and Notification System" (Accepted)

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